

ELK Stack

ELK is a collection of three open-source tools used for **log management, monitoring, and Security Information and Event Management (SIEM)**.

ELK = Elasticsearch + Logstash + Kibana

1. Elasticsearch

- Stores and indexes log data.
- Provides fast searching and analysis of large datasets.

Example: Stores SSH login logs, system logs, and firewall events.

2. Logstash

- Collects logs from various sources.
- Parses, filters, and transforms log data.
- Sends processed logs to Elasticsearch.

Example: Receives logs from Filebeat and forwards them to Elasticsearch.

3. Kibana

- Visualizes data stored in Elasticsearch.
- Creates dashboards, charts, and reports.
- Supports searching and monitoring of events.

Example: Displays failed SSH login attempts and network activity.

Data Flow

Log Sources

Filebeat/Elastic Agent

Logstash

Elasticsearch

Kibana

Main Uses

- Centralized log collection.
- Security monitoring (SIEM).
- Application and server monitoring.
- Network troubleshooting.
- Compliance and auditing.
- Threat detection and incident investigation.

“ELK is a log management and SIEM platform where Logstash collects and processes logs, Elasticsearch stores and indexes them, and Kibana provides visualization and analysis through dashboards.”